

# TFPT — Frontier items

$\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and quantum gravity —  
the honest status, not forced onto the ladder

Stefan Hamann

Alessandro Rizzo

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## What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of  $\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading document 4 (Frontier).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed (**[E]**); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status                  |
|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive <b>[O]</b>    |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | <b>[E]</b> / <b>[C]</b> |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | <b>[C]</b>              |

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**[E]** exact / machine-proven

**[C]** conditional

**[O]** open / axiom

**[X]** kill test


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**Ground rule.** These five are exactly the items previously flagged “*not on a clean action/ladder yet — do not force.*” This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

#### Hard rule on the frontier — one interface $F_{\text{transfer}}$

**Koide,  $\eta_B$ , the axion relic scale and  $m_p/m_e$  are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied.** They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor*  $F_{\text{transfer}}$  — the continuous transport from compiler *source* data to measured observables (Koide source→pole;  $\eta_B$  source→Boltzmann relic; axion scale→cosmological abundance;  $m_p/m_e \rightarrow$  QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named, not-yet-performed calculation. In particular  $\eta_B$  is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

*Why this is principled, not a gap:* these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [verification/v35\_quark\_qcd\_boundary.py]

*Machine-enforced.* This rule is a CI-style guard ([verification/v187\_ftransfer\_laws.py]): the four frontier families ( $F_{\text{pole}}$  Koide,  $F_{\text{Boltzmann}}$   $\eta_B$ ,  $F_{\text{relic}}$  axion,  $F_{\text{QCD}}$   $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — exact algebraic sub-parts (the Koide 53/54 factor,  $b_3 = -7$ ) may be [E], but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([verification/v188\_frontier\_wording\_guard.py]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

*The functor contract CONTRACT.F.01.* Beyond the typing guard, the functor is pinned by four checkable structural axioms ([verification/v213\_ftransfer\_functor.py]): each instance is a *discrete* compiler kernel [E] composed with an *external* continuous solver [C], and the functor is (1)  $\mu_4$ -deck equivariant (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue), (2) Plücker-preserving ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\|\text{Pl}(K)\|_1 = 11$ ), (3) positive/stochastic ( $\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ ,  $\kappa_f \in (0, 1)$ ), and (4) external-module-explicit ( $b_3 = -7$  a carrier output, the Boltzmann/relic/QCD content named and external). So  $F_{\text{transfer}}$  is a typed functor, not a bag of open topics — the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), and a consolidation, not a closure.

## 1 Baryon asymmetry $\eta_B$ — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

**Route A — from the closed baryon fraction.** The seed already fixes the baryon *fraction* (the  $E_8$  decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \frac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \quad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed  $\Omega_b \approx 0.049$ . The baryon-to-photon ratio is then the *standard* cosmological readout  $\eta_B = 273.9 \times 10^{-10} \Omega_b h^2$ . With the TFPT Hubble value ( $h \approx 0.674$ , from  $H_0 \sim \sqrt{\Lambda}$ ),

$$\boxed{\Omega_b h^2 = 0.0222, \quad \eta_B = 6.09 \times 10^{-10}} \quad (\text{observed } 6.1 \times 10^{-10}).$$

So  $\eta_B$  is a *downstream cosmological readout* from the closed  $\Omega_b h^2$  handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

**Route B — leptogenesis interface ([C]).** The TFPT branch fixes the neutrino spectrum, the CP phase structure ( $\delta_{\text{CKM}}$ ,  $\delta_{\text{CP}}^\nu$ , both closed) and a *plausible washout anchor*  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV. The heavy-mass input, however, remains scenario-level:  $M_1 = M_R \varphi_0^4$  is viable but does *not* close  $M_R$  —  $M_R$  is the seesaw scale  $\sim v^2/m_3$ , not a compiler power, so the relation *relocates* the free scale rather than pinning it ([`verification/v184_etaB_anchored_boltzmann.py`]). The numerical  $\eta_B$  then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero. **Honest caveat:** that Boltzmann network (the washout/efficiency  $\kappa_f$ , the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed  $\Omega_b h^2$ ) is quoted.

**The interface, operationalised ([C]).** The Boltzmann path can be made falsifiable: with the standard estimate  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted  $\delta_{\text{CP}}^\nu = 4\pi/3$  — the Davidson–Ibarra asymmetry  $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3 / v^2$  and the washout efficiency  $\kappa_f(\tilde{m}_1)$  give, over the natural ranges  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and  $\tilde{m}_1 \in [m_\star, m_{\text{atm}}]$ , an  $\eta_B$  band  $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$  that *brackets* the observed  $6.1 \times 10^{-10}$  (a canonical point  $M_1 = 10^{10}$  GeV,  $\tilde{m}_1 = 5 \times 10^{-3}$  eV gives  $6.0 \times 10^{-10}$ , untuned). So the route is *viable*; but  $M_1$  and the washout are scenario inputs, so  $\eta_B$  stays [C] — if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([`verification/v169_etaB_boltzmann_interface.py`]).

#### Honest status of $\eta_B$ — strictly typed [C]

The two statements must be kept apart and not interchanged:

$\eta_B$  as a cosmological readout from the closed  $\Omega_b h^2$  is *determined* ( $6.1 \times 10^{-10}$ )

$\eta_B$  as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So  $\eta_B$  is [C] — determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

#### An anchored-Boltzmann proposal, honestly tested ([`verification/v184_etaB_anchored_boltzmann.py`]).

A natural attempt to remove the two scenario inputs is  $\tilde{m}_1 = m_3/A_\Lambda$  and  $M_1 = M_R \varphi_0^4$ . The first *works* as a [C] anchor:  $\tilde{m}_1 = m_3/10 \approx 5 \times 10^{-3}$  eV ( $A_\Lambda = 10 = |E(K_5)|$  the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*:  $M_R \sim v^2/m_3$  is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of  $\bar{M}_{\text{Pl}}$  both miss it (the  $c_3$ - and  $\varphi_0$ -powers are off by  $\sim 75\%/ \sim 40\%$ ), so  $M_1 = M_R \varphi_0^4$  merely *relocates* the free input  $M_1 \rightarrow M_R$ . Feeding both into the standard estimate lands  $\eta_B$  in the observed band  $O(10^{-10})$ , so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero.  $\eta_B$  **stays [C], a sharper scenario, not a derivation.**

**A cleaner route: the scalaron-decouple branch** ([`verification/v212_leptogenesis_decouple.py`]). Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decouple  $A_\Lambda = 10 = |E(K_5)|$ :

$$M_1 = \frac{M_{\text{scal}} (\varphi_0^{\text{ret}})^2}{A_\Lambda} \approx 8.65 \times 10^9 \text{ GeV}, \quad \tilde{m}_1 = \frac{m_3}{A_\Lambda} \approx 5 \text{ meV}.$$

$M_1$  now uses *only* compiler quantities ( $M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}}$ ,  $\varphi_0^{\text{ret}}$ ,  $A_\Lambda$ ) — *no hidden seesaw scale* — and lands inside the thermal window  $[3 \times 10^9, 3 \times 10^{10}]$  GeV where the Davidson–Ibarra  $\varepsilon_1 \approx 8.5 \times 10^{-7}$  and the strong-washout efficiency give  $\eta_B \approx 1.2 \times$

$10^{-9}$ , bracketing the observed  $6.1 \times 10^{-10}$  untuned. The *full* BDP Boltzmann ODE (experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py) now confirms this: the integrated efficiency  $\kappa_f = 0.092$  (validating the analytic  $\kappa_f$ ) gives  $\eta_B = 6.5 \times 10^{-10}$  at the frozen  $M_1$  — a factor 1.07 from the observed value, with *no* free  $M_R$  dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing  $A_\Lambda$  reads as *scalaron reheating dressed by two seed insertions*  $(\varphi_0^{\text{ret}})^2$ , *over the pair sector*. This is cleaner than  $M_1 = M_R \varphi_0^4$  (no relocated free scale), but the coupling *mechanism* is posited, not derived, and  $\kappa_f$  is washout-dependent — so  $\eta_B$  stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

## 2 Proton/electron ratio $m_p/m_e$ — a cross-sector ratio

$m_e$  is *closed* on the  $\varphi_0^{\text{ret}}$ -ladder ( $\hat{m}_e = \frac{v_{\text{geo}} \pi}{\sqrt{2}} \frac{16}{7} (\varphi_0^{\text{ret}})^5$ ). The proton mass is *not* a Yukawa: it is QCD-dominated,  $m_p \sim \text{a few} \times \Lambda_{\text{QCD}}$ , and  $\Lambda_{\text{QCD}}$  arises by dimensional transmutation from  $\alpha_s$  running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single  $\varphi_0^{\text{ret}}$  power:  $1/(\varphi_0^{\text{ret}})^2 = 353.7$  and  $1/(\varphi_0^{\text{ret}})^3 = 6651$  bracket 1836 with no clean rung in between.

The numerator is *parameter-free*, the ratio is *not forced*. Running  $\alpha_s(M_Z)$  down with the carrier coefficient  $b_3 = -7$  (the [verification/v159\_pyrat\_gauge\_crosscheck.py] value, two-loop +  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and reaches confinement ( $\alpha_s \sim 1$ ) at  $\sim 0.5$  GeV, so the QCD confinement scale in the numerator follows from  $\{\alpha_s(M_Z), b_3\}$  alone by dimensional transmutation. The  $O(1)$  proton/ $\Lambda$  factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([verification/v164\_qcd\_scale.py]).

The  $F_{\text{QCD}}$  transfer is now an implemented solver ([verification/v262\_fqcd\_mp\_me.py]). The contract-only pipeline (experiments/ftransfer/qcd\_matching\_mp\_me) is realised: a two-loop  $\alpha_s$  run from  $M_Z$  with  $n_f$  thresholds (carrier coefficient  $b_3 = -7$ , [E]) gives  $\alpha_s(2 \text{ GeV}) \simeq 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \simeq 0.33$  GeV; with the lattice bridge  $C_p = m_p/\Lambda^{(3)}$  (external  $O(1)$ ) and the closed  $m_e$  this reproduces  $m_p/m_e \simeq 1824$  against the measured 1836.15 within the external error budget ( $\sim 7\%$ , lattice- $C_p$  dominated). This advances FR.MPME.01 ([O]) to FR.MPME.02 ([C], “reproduced by the named  $F_{\text{QCD}}$  transfer”); the firewall holds —  $b_3 = -7$  is [E], the ratio is a transfer, never a compiler power.

### Honest status of $m_p/m_e$ [O]

Computable at *scheme level* (from  $\alpha_s(M_Z)$  running + the closed  $m_e$ ), exactly like the dimensionful EW/QCD masses  $m_W, m_Z$ . But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

**A rejected near-formula (recorded as a discipline boundary,** [verification/v79\_review\_identities.py]). The combination  $|W(D_5)| - \dim \Lambda^3(SU(9)) + N_{\text{fam}} \lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$  matches the PDG value to  $\sim 10^{-6}$  — and is *deliberately rejected*. Two reasons make it numerology, not a derivation:  $SU(9) = A_8$  (the  $\dim \Lambda^3 = 84$  block) does *not* occur in the carrier  $D_5 \oplus A_3$ , and  $m_p$  is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it;  $m_p/m_e$  stays [O]. (The same firewall rejects  $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q} c_3^6 \approx 6.10 \times 10^{-10}$ : arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output —  $\eta_B$  stays [C], the downstream readout above.)

## 3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral  $c=8$  fixed point ([verification/v158\_fixed\_point\_stable.py]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling  $\lambda$  is the one marginal scalar

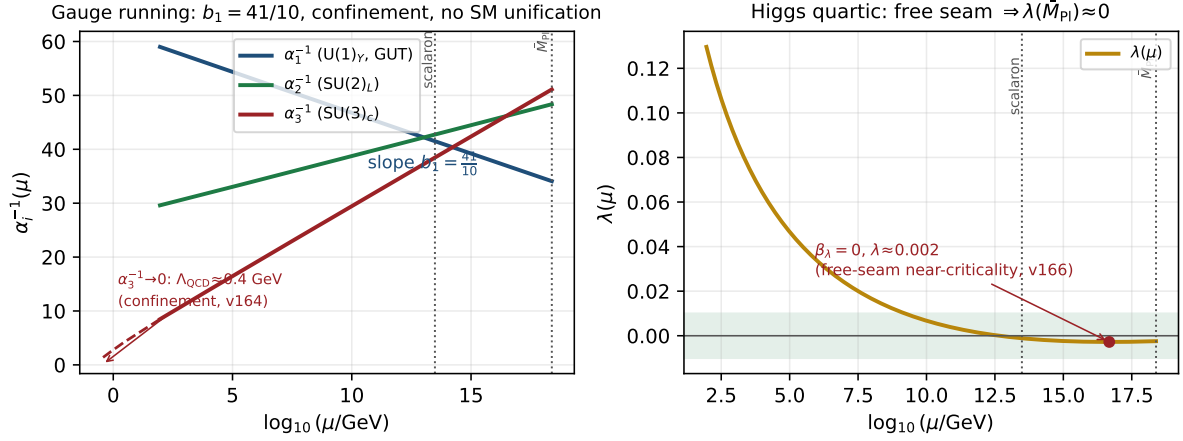


Figure 1: The  $F_{\text{transfer}}$  running (two-loop SM RGEs; the gauge  $\beta$ -functions are confirmed by PyR@TE, [verification/v159\_pyrate\_gauge\_crosscheck.py]). *Left*: the gauge couplings — the  $U(1)_Y$  slope is  $b_1 = 41/10$ ,  $\alpha_3^{-1} \rightarrow 0$  at  $\Lambda_{\text{QCD}} \approx 0.4$  GeV (confinement, [verification/v164\_qcd\_scale.py]), and the Standard Model does *not* unify. *Right*: the Higgs quartic falls to  $\lambda(M_{\text{Pl}}) \approx 0.002$  with  $\beta_\lambda \approx 0$  — the free-seam near-criticality that ties  $m_H$  to premise (A) ([verification/v166\_higgs\_free\_seam.py]).

coupling, so a free seam forces it to the Gaussian point there:

$$\lambda(M_{\text{seam}}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\text{seam}}) = 0$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from  $M_Z$  upward with the measured  $(m_H, m_t)$  gives  $\lambda(M_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(M_{\text{Pl}}) \approx 0$  at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate  $\sim 0.022$ ). Imposing the double condition predicts  $m_H \sim 129\text{--}134$  GeV (a band, sensitive to  $m_t$  and  $\alpha_s$ ); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ( $c_3^{7/2} M_{\text{Pl}} \sim 3 \times 10^{13}$  GeV) gives  $m_H \sim 107$  GeV (too low), so the boundary condition lives at  $M_{\text{Pl}}$ , consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only  $N_\Phi = 1$ ); it is typed [E] (the freeness signature and scale selection) and [C] (the  $m_H$  band) ([verification/v166\_higgs\_free\_seam.py]). Figure 1 shows the whole  $F_{\text{transfer}}$  running.

## 4 The Koide relation — near $2/3$ , computed exactly

With the closed lepton source masses  $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$  the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\dots, \quad \frac{2}{3} = 0.66667, \quad \text{deviation} \quad -0.33\%.$$

The measured pole-mass value is famously  $Q_{\text{PDG}} = 0.66666$ . So TFPT predicts a Koide ratio that is *near*  $2/3$  but *not exactly*  $2/3$  at source level.

### Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$  is a *prediction*, 0.33% below  $2/3$ . Exact  $2/3$  would require the three carrier rationals  $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$  to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force  $2/3$ .

**The *target*  $2/3$  has a compiler reading  $|\mathbb{Z}_2|/N_{\text{fam}}$ .** The  $E_8$ -slice scan adds the missing piece: the value the data sits on,  $Q = 2/3$ , is exactly  $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ , the sheet over the family count — the same two atoms  $(2, 3)$  that run through the whole compiler. So the picture splits cleanly: the *democratic target* is  $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$  (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus “democratic  $2/3$  from  $(|\mathbb{Z}_2|, N_{\text{fam}})$ , realised to 0.33% by the explicit masses” — promoted from a bare near-miss to a reading with a target. [C]

**Conjecture: a source→pole correction of size  $\varphi_0/|W(A_3)|$  ([C]).** The source value sits just below  $2/3$ , and the deficit is almost exactly the family-Weyl quantum  $\varphi_0/|W(A_3)|$  with  $|W(A_3)| = 4! = 24$ :

$$Q_\ell^{\text{source}} = 0.6644638, \quad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting  $\frac{2}{3} = 0.6666667$  by only  $1.3 \times 10^{-5}$ . So the *conjecture*  $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$  would land essentially on the democratic  $2/3$ . Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [verification/v25\_frontier\_conjectures.py]

**Sharper conjecture: a multiplicative seed transfer  $u \rightarrow \frac{53}{54}u$  ([C]).** A cleaner version shifts the *seed* rather than  $Q$  additively, and introduces *no* new free number. Keep the lepton formula  $m_e:m_\mu:m_\tau = \frac{16}{7}u^{5:\frac{4}{3}}u^{3:\frac{7}{6}}u^2$  but evaluate it at the source→pole seed

$$u_{\text{pole}}^{(\ell)} = \left(1 - \frac{1}{2N_{\text{fam}}^3}\right)u = \frac{53}{54}u, \quad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2| N_{\text{fam}}^3.$$

Then  $Q_\ell(\frac{53}{54}u) = 0.66666610$ , i.e.  $2/3$  to within  $-5.7 \times 10^{-7}$  — an order of magnitude tighter than the additive  $\varphi_0/24$  correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is  $u \mapsto u(1 - \frac{1}{2N_{\text{fam}}^3})$*  (a renormalised-seed effect from the leptonic self-energy or the  $A_3$  family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37\_plucker\_anchor.py]

**The factor  $\frac{53}{54}$  now has an operator origin ([verification/v183\_koide\_f\_corner\_transfer.py]).** It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner  $F = R + Q = M(1, 1)$  over the sheet-doubled  $E_6 \times A_2$  cubic family block,

$$\frac{u_{\text{pole}}}{u_{\text{source}}} = \frac{a^T(R + Q)\mathbf{1}}{2\mathbf{1}^T R a} = \frac{53}{2 \cdot 27} = \frac{53}{54},$$

where  $\mathbf{1}^T R a = 27$  is the cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ ) and  $a^T(R + Q)\mathbf{1} = 53 = 52 + 1$  with  $\det B_F = 52 = \dim F_4$  (v80). A hard negative control fixes the corner:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. So the *factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the  $F$  corner — an  $F_{\text{transfer}}$  special case ( $F_{\text{pole}}$ ). Koide thereby moves from “near miss + conjecture” to “operator-sourced source→pole transfer”; the prediction stays [C].

**The Koide target and the QFT transfer gap are the *same* quotient ([E]/[C]).** The same  $Q_\star$  controls a second, independent object — the first non-trivial eigenvalue of the admissible  $C_6$  transport operator. Its spectrum is  $\{1, (2/3)^6, (1/3)^6\}$ , so

$$\lambda_2 = \left(\frac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \quad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6.$$

The Koide democratic target, the QFT mass-gap eigenvalue  $\lambda_2$  (hence  $\Delta = 6 \log \frac{3}{2} = -\log \lambda_2$ ), and the  $A_3$  positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not “solve” Koide, but it

explains *why* the near-value  $2/3$  recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

**The source→pole map is forced by that gap ([E]/[C]).** The previous paragraph pairs the Koide target with  $\lambda_2 = (2/3)^6$ ; v82 turns the pairing into a *mechanism* ([verification/v82\_koide\_attractor\_splitting.py]). On the anchor double cover (the flavor sector) the two branch points are Koide  $q = 2$  and carrier  $q = 5$ ; any branch-preserving Möbius transfer that fixes both is, in  $\rho = (q - 2)/(5 - q)$ , exactly  $\rho \mapsto \mu\rho$ , hence *unique* once  $\mu$  is set. Choosing  $\mu = \lambda_2 = (2/3)^6$  (no new number) gives the unique map  $F(q) = 2(569q - 3325)/(665q - 3517)$  with  $F'(2) = (2/3)^6 < 1$  and  $F'(5) = (3/2)^6 > 1$ : **Koide is the attractor, the carrier the repeller**, and  $q_n \rightarrow 2$  ( $x \rightarrow -2/3$ ) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value  $2/3$ , the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture  $u \mapsto (53/54)u$  above; both contract the source onto  $2/3$ .)

**Relaxation toy: basin, exact rate, and two honest negatives ([E]).** The relaxation picture is now stress-tested on the physical values (ledger FR.KOIDE.05, [verification/v93\_koide\_relaxation\_toy.py]).

**Basin lemma [E] (new):** under the canonical dictionary  $q = N_{\text{fam}} Q$  the *entire* physical Koide range  $Q \in [\frac{1}{3}, 1]$  (Cauchy–Schwarz) maps to  $q \in [1, 3]$  — the attractor  $q = 2$  lies inside, the repeller  $q = 5$  outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly*  $(2/3)^6$  per step. **Source/pole [E]:** the exact ladder gives  $Q_{\text{src}} = 0.6644638$ , i.e.  $\rho_{\text{src}} = -0.0021980 \approx -\varphi_0/24$  to 0.8% — the  $\varphi_0/24$  conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ( $24 = |W(A_3)|$ ); PDG pole masses give  $Q_{\text{pole}} = 0.6666645$ , the branch point hit to  $3.3 \times 10^{-6}$ . **Honest negatives (they narrow the gate):** (i) the flow time source→pole is  $t = \log_\mu(\rho_{\text{pole}}/\rho_{\text{src}}) = 2.84$  — *not* an integer, so the literal discrete iteration  $\text{pole} = F^n(\text{source})$  is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator  $\exp(t \log F)$ ; (ii) the 0.79% mismatch of  $\rho_{\text{src}}$  vs  $-\varphi_0/24$  is far above ladder precision, so “exactly  $\varphi_0/24$ ” remains a conjecture, not an identity. The open question is no longer “why  $2/3$ ?” but “what is the continuous transfer generator, and why is its flow time  $\sim 2.8$  periods?”

**Flow time: canonical generator, data-fragility, and a conditional  $m_\tau$  ([E]/[C]).** Three sharpenings of the paragraph above (ledger FR.KOIDE.06, [verification/v99\_koide\_flow\_time.py]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\frac{dq}{dt} = \frac{\Delta}{N_{\text{fam}}} (q - 2)(q - 5) = \frac{\Delta}{N_{\text{fam}}} \det B(q), \quad \Delta = 6 \log \frac{3}{2} \quad (\text{the gap}),$$

whose time-1 map *is* the Möbius map  $F$  (machine-checked;  $e^{-\Delta} = (2/3)^6$  exactly). The missing [C] object is therefore precisely a QFT mechanism producing *gap*  $\times$  *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of  $t$  is data-fragile [E].** Because  $\rho_{\text{pole}} = -2.2 \times 10^{-6}$  sits so close to the branch point,  $t$  is hypersensitive to  $m_\tau$  (PDG  $1776.93 \pm 0.09$  MeV):  $t(-1\sigma) = 2.35$ ,  $t(\text{central}) = 2.84$ , and  $Q$  crosses  $\frac{2}{3}$  *inside* the error band (at  $+0.43\sigma$ , where  $t \rightarrow \infty$ ) — within  $+0.5\sigma$ ,  $t$  passes *every* integer  $\geq 3$ . So the exclusion of  $\text{pole} = F^n(\text{source})$  holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):**  $n=2$  is *excluded* ( $-2.9\sigma$ , [E]);  $n=3=N_{\text{fam}}$  steps — one per family — predicts

$$m_\tau = 1776.9427 \text{ MeV} \quad (+0.14\sigma; \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while  $n=4$  (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is  $\sigma(m_\tau) \sim 0.01$  MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading  $t =$

$\ln(M_{\text{scal}}/m_\tau)/\ln W$  matched  $W = 46080 = |W(D_5)||W(A_3)|$  to  $10^{-4}$  — and is *destroyed* by negative controls (the same hypersensitivity makes *any*  $W$  in a wide range land within  $0.1\sigma$ ); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

## 5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Two of its inputs are already *closed*:

$$\text{initial misalignment } \theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_*)) = 2.974 \text{ rad} = 170.4^\circ,$$

and the domain-wall number  $N_{\text{DW}}$  winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\text{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\text{str}}(f_a, N_{\text{DW}}),$$

and the minimal closed branch therefore has *dominant axion-like dark matter*.

**The  $E_8$  scan rules out a rival WIMP candidate.** The  $D_8 = SO(16)$  slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor  $128 = (16, 4) \oplus (\overline{16}, \overline{4})$  is the *same* matter spinor as in the  $D_5 \times A_3$  construction — there is *no spare  $E_8$  singlet* left over to play a WIMP. Every  $E_8$  direction is accounted for by the carrier  $\times$  family content. So dark matter cannot be a new  $E_8$  representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E]

### Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion) and its misalignment angle  $\theta_i = 170^\circ$  is *closed* [E]. The relic abundance still needs the decay constant  $f_a$  and mass  $m_a$  (interface data set by the  $M_R/\Lambda$  scales) — a standard axion-DM computation, not yet a compiler power. So DM is “candidate + misalignment fixed, scale  $f_a$  pending” — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near  $M_R$ .)

**Conjecture: a determinant-line axion scale**  $f_a = M_{\text{scal}}/128$  ([C]). A natural compiler scale for the decay constant is the scalaron mass over the full carrier  $\times$  deck multiplicity,

$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ |\mu_4|} = \frac{M_{\text{scal}}}{128}, \quad M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}} \approx 3.06 \times 10^{13} \text{ GeV},$$

giving  $f_a \approx 2.39 \times 10^{11} \text{ GeV}$  and, via the standard QCD-axion relation  $m_a \simeq 5.70 \mu\text{eV} (10^{12} \text{ GeV}/f_a)$ , a mass  $m_a \approx 23.8 \mu\text{eV}$ . This is consistent with the closed near-hilltop misalignment  $\theta_i = 170.4^\circ$ , but it is a *cosmological scenario* (sensitive to anharmonic / string corrections near the hilltop), not a closed hit. [verification/v25\_frontier\_conjectures.py]

**The haloscope coupling is tied to  $c_3$ .** In the determinant-line normalization (Lagrangian  $\mathcal{L} \supset c a F \tilde{F}$ ) the axion–photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \quad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and the cosmic birefringence  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$ , which still carries  $f_a$ ). The small-angle reading  $\theta_i = \varphi_0$  of the determinant-line axion (an earlier note) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$  above; it is recorded as a retired reading, never used.

**Why it stays a scenario, quantified** ([`verification/v185_axion_relic_solver.py`]). The harmonic misalignment *under-produces* at this  $f_a$ :  $\Omega_a h^2 \sim 0.12 (f_a/9 \times 10^{11})^{7/6} \langle \theta^2 \rangle$  gives a prefactor  $\sim 0.026$  per  $\theta^2$ , so  $\theta_i \sim O(1)$  would make only  $\sim 21\%$  of  $\Omega_{\text{DM}}$ ; reaching the observed  $\Omega_{\text{DM}} h^2 = 0.12$  *requires*  $\theta_{\text{eff}}^2 \sim 4.7$ , i.e. exactly the large-angle (anharmonic) enhancement at the  $\theta_i \approx 170^\circ$  hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to  $\theta_i$  and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract  $F_{\text{relic}}(f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ, N_{\text{DW}}=1) \stackrel{?}{=} \Omega_{\text{DM}}$  is a kill test for the *branch*, not the theory.

**A full finite- $T$  solve confirms an over-production tension.** Both a quick  $\theta^2 F$  estimate ( $F(\theta_i) \approx 4$  at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + (3 + \frac{d \ln H}{dN}) \theta' + (m_a(T)/H)^2 \sin \theta = 0$$

(`axion_relic/full_finiteT_solve.py`: lattice susceptibility  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ , converged and normalised so  $\theta_i=1$  gives the standard  $\Omega_a h^2 \approx 0.03$ ) agree: at the *predicted*  $\theta_i \approx 170^\circ$  the relic is  $\Omega_a h^2 \approx 0.66$ ,  $\sim 5.5\times$  above  $\Omega_{\text{DM}} h^2 = 0.12$ . The observed value is reached only at  $\theta_i \approx 106^\circ$ , not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at ( $f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ$ ) *over-closes* the universe unless there is extra dilution (entropy injection) or a lower  $f_a$ . This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

**A more robust angle: the spine branch.** [`verification/v211_axion_spine_angle.py`] The same finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\theta_i \approx 106^\circ$  — and the central spine quotient gives  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$  *exactly* (a rational multiple of  $\pi$ , no fit; harmonic estimate  $0.03 \theta_i^2 = 0.107$ , lifted to  $\sim 0.12$  by a mild anharmonic factor). Crucially  $108^\circ$  lies  $62^\circ$  *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma) \approx 170^\circ$  (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (DM.AXION.SPINE.01), *not* a derivation; [X] a converged  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  at  $\theta_i = 3\pi/5$  demotes the branch.

## 6 The muon anomalous magnetic moment — a seam vertex read-out

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes  $\alpha$ : with  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4 = 48c_3^4 = 3/(256\pi^4)$  and the carrier compression quotient  $B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}$  ([`verification/v51_boundary_half_step.py`]),  $\delta_2 = B\gamma \delta_{\text{top}}^2 = \frac{5}{4} \delta_{\text{top}}^2$ . Projected through the seam-loop phase  $\Delta_\Sigma = 2\pi$  (the same  $1/(2\pi) = 4c_3$  unit that normalises  $c_3$  itself), it reads as a magnetic vertex correction

$$a_\mu^{\text{seam}} = \frac{\delta_2}{2\pi} = \frac{45}{524288 \pi^9} \approx 2.879 \times 10^{-9} \quad [\text{C}].$$

The *value* is an exact compiler number (only  $c_3, \Omega_{\text{adm}}, B\gamma$ ; trace reading  $\delta_2 = 4! \text{Tr}_{S^+}(X^2) c_3^8$ ,  $\text{Tr} = 120 = 5!$ ) — but the *identification* of  $\delta_2/(2\pi)$  as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [`verification/v204_muon_seam_g2.py`]

**Data, honestly.** Against the dispersive discrepancy  $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$  (Fermilab/BNL world average minus the WP2020 SM) this is  $0.81\sigma$ . **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ( $\Delta a_\mu \sim 1.5 \times 10^{-9}$ ); the TFPT value is *fixed* and would then sit  $\sim 1.5\sigma$  high. A converged  $\Delta a_\mu$  outside  $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$  excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

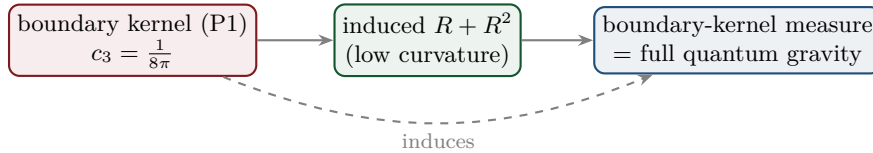
## 7 Full quantum gravity — induced from the boundary kernel

*Framing (read this first):* full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through  $v_{\text{geo}}$  and  $F_{\text{transfer}}$  and are falsifiable now (JUNO, CMB-S4, EDM).  $G_{\text{net}}$  certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation  $c_3 = 1/(8\pi)$  is the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The  $R^2$ /scalon sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton’s constant is then not an independent input:* fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{P1}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* selecting the physical  $\Lambda$  branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60\_lambda\_metrology\_branch.py]) — a metrology readout, not a from-nothing prediction.

**Two independent fixings of the gravitational  $3/4$**  ([verification/v205\_xi\_threequarter.py]). The induced-gravity coefficient the seam-determinant replica pins,  $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$  ([verification/v152\_norm\_is\_anchor.py]), is reached a *second*, independent way: the old torsion-compression ansatz  $\kappa^2 = \xi \varphi_0^{\text{ret}}/c_3^2$  with the Einstein limit  $\kappa^2 = 8\pi G$  forces  $\xi = c_3/\varphi_0^{\text{ret}}$ , which at tree level  $\varphi_0^{\text{ret}} \rightarrow 1/(6\pi)$  is *exactly*  $3/4$  (with the retained seed 0.7483,  $-0.22\%$ ). So  $3/4$  appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and  $1/G$  stays the one declared anchor.

**The  $\Lambda$  branch sharpens  $H_0$**  ([verification/v206\_h0\_lambda\_branch.py]). Reading the seam vacuum number  $\delta_\Sigma = \Omega_{\text{adm}} e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$  (lowest admissible block  $m_1 = \Omega_{\text{adm}} = 48$ ) through the flatness closure gives a TFPT Hubble value  $H_0 = 66.5\text{--}67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $1.6\text{--}0.5\sigma$  below Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and  $H_0$  uses the imported  $\Omega_b, \omega_{\text{DM}}$ , so it is a [C] readout;  $|\log_{10} \delta_\Sigma| \approx 122.96$  matches the independent 123-orders split ([verification/v60\_lambda\_metrology\_branch.py]), and  $M_{\text{P1}}^4/\bar{M}_{\text{P1}}^4 = (8\pi)^2$  must be displayed, never silently interchanged.



### Honest status of quantum gravity — physical sector closed, ambient gap-decoupled [E]/[O]

“Full quantum gravity beyond the  $R^2$  sector” is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action  $\text{Tr } f(D/\Lambda)$  gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ( $a_2 \sim -R/3$ ,  $a_4|_{R^2} \sim R^2/72$ ), Einstein–Hilbert +  $R^2$  *structurally* — the  $R + R^2$  form is not postulated; the scalaron ratio  $M^2/\bar{M}_{\text{P1}}^2 = 6(4\pi)^2/f_0$  with the TFPT closure  $c_3^7$  fixing the cutoff moment  $f_0$ . *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([verification/v255\_spectral\_action\_expansion.py]), and the cutoff moment  $f_0$  is no longer a free scheme choice: identifying  $f$  with the *seam KMS weight* ( $\beta=1$  by Tomita–Takesaki,  $2\pi=1/(4c_3)$ ) gives  $f_2/f_0 = 1$  exactly ([verification/v259\_modular\_cutoff\_kappa.py]), so  $\kappa = \sqrt{(f_2/f_0)} c_{\text{PS}}/c_{\text{grav}}$  collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a strictly positive effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the un-built ambient, so under those assumptions the admissible-sector measure is closed. **What remains [O]:** the ambient metric measure — the projective limit (G6) — is the  $G_{\text{metric}}$  research gate; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36\_spectral\_action\_g2.py]

*Update (the QG.AMB.01 roadmap, consolidated — [verification/v275\_qgamb\_roadmap.py]).* The open gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 2 \cdot 248 c_3^2 = 1.648 > 0$ ) makes every low-energy readout independent of the un-built measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic  $(E_8)_1$  net ( $c=248/31=8$ , v77), reduced not closed; the *perturbative* 4d layer is now in hand (v269/v271/v273). [O] what remains is the one genuine structural frontier — the nonperturbative interacting ambient measure (the bulk projective limit), which [X] closes iff the seam net  $= (E_8)_1$  and the bulk reconstruction is proved (QGAMB.ROADMAP.01).

*Tier B made concrete ([verification/v277\_seam\_calderon\_e8\_match.py]).* The seam-Calderón  $\rightarrow (E_8)_1$  identification is reduced to a *single* bit. [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$  (Sugawara 248/31),  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248 = \dim E_8$  weight-1 currents. [E] the discriminator is *holomorphy*: the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (four primaries, non-holomorphic) — same  $c$ , different det, so the one discriminating condition is  $|\det K|=1$ . [C] the seam side matches via the flat  $\tau=i$  pillowcase (v276) and the Kummer/K3 (v260). [O] the residual is exactly that single holomorphy bit — reduced, not closed (QGAMB.TIERB.01).

*The modular-data view ([verification/v281\_holomorphy\_modular\_data.py]).* The Tier-B bit is the anyon count:  $\# \text{anyons} = |\det \text{Gram}|$  ( $E_8 \rightarrow 1$  holomorphic,  $SO(16) \rightarrow 4$ ), the condensation tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  each step a Lagrangian  $Z_2$  boson, and Gauss–Milgram returns  $c=8$  — so the open bit is simply “does the seam net condense to  $\det K=1$ ” (QGAMB.MODULAR.01).

**The live kill-test scorecard ([verification/v283\_live\_killtest\_scorecard.py]).** The recent round’s computable predictions confronted with current data:  $\sin^2 \theta_{12}$  ( $+0.31\sigma$ ),  $\sin^2 \theta_{13}$  ( $+1.45\sigma$ ),  $J_{\text{PMNS}} = -0.0297$ ,  $\Sigma m_\nu$  floor 0.0586 eV, proton decay  $\tau_p = 1.55 \times 10^{35}$  yr — all consistent at  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$ : DESI/CMB-S4; proton decay: Hyper-K) (PRED.LIVE.01).

*Update (QBL chain, v108–v113):* the boundary-net residual now carries *double* weight — the carrier choice  $g = 5$  has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank =  $c$ : 5 carrier block, 8 seam hull) is the defining datum of the free  $c = 8$  net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection [E] (see `tfpt_1` and `tfpt_research_contracts`). [verification/v113\_quasifree\_kernel.py]

*Update (free-bulk premise discharged, v160–v165):* that “free  $c=8$  net” premise is no longer free-standing. It is a fixed-point theorem (quasi-free  $\Rightarrow$  all higher cumulants  $\kappa_{2n}=0$ , v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap  $(2/3)^6$  (v161), forced by  $\mu_4$ -equivariance (v162); and the residual factors into the already-open  $A_2$  net-existence (an assembly of established net constructions, [C]) and  $R_1$  is the seam-realisation keystone SEAM.EQUIV.01, with the irreducible core  $\{\pi, v_{\text{geo}}\}$  a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [verification/v165\_premise\_a\_closure.py]

*The deduction below the premise is Lean-formalised (FORM.QGEO.02/FORM.QGEO.01).* After the residual reduces to the single seam-realisation postulate QGEO.SYM.01 (the carrier  $\mu_4$  clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the UNIFORM node:  $z \mapsto iz$ ,  $\sigma\rho\sigma = \rho^{-1}$ ,  $\text{orbit} \rightarrow \mu_4$ , multiplier order-4  $\Leftrightarrow \zeta = \pm i$ ) and the conditional implication  $\text{mark-local} \Rightarrow \omega \circ \rho = \omega$  are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors [verification/v177\_seam\_marking\_kernel.py] and [verification/v210\_mark\_local\_dtn.py]). So everything *downstream* of the postulate is [E]; QGEO.SYM.01 *itself* stays the one [O] bedrock.

*A second, independent route ([verification/v207\_asymptotic\_safety.py], [O]).* Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point —  $\partial_t g = (2 + \eta_R)g$  is UV-repulsive at  $g \rightarrow 0$  and damped at large  $g$  by the positive-definite torsion  $K^2$  term, so  $2 + \eta_R = 0$  has a root  $g_* > 0$  (intermediate value), with the scale-invariant axion coupling fixed and small,  $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$ . This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close  $G_{\text{metric}}$ ; the primary, machine-checked route stays the seam-net.

### The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input ( $v_{\text{geo}}$  = the reduced Planck mass  $M_{\text{Pl}}$ ) is not a hidden gap but the *unit*, and it is over-determined ([`verification/v274_scale_overdetermination.py`]). [E] every dimensionful output is a pure number  $\times M_{\text{Pl}}$ :  $\rho_{\Lambda}/M_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ ,  $M_{\text{scal}}/M_{\text{Pl}} = c_3^{7/2}$ , the whole spectrum — the absolute  $\nu$ -scale (v272) and  $v_{\text{geo}}$  are the *same* anchor. [C] **two independent routes agree**: gravity gives  $M_{\text{Pl}} = (8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV, and inverting the compiler's dark-energy prediction with the measured  $\rho_{\Lambda}^{1/4} = 2.24$  meV gives  $M_{\text{Pl}} = 2.438 \times 10^{18}$  GeV — the same Planck mass to 0.11% (*conditional* on the  $\Lambda$ -branch readout LAMBDA.BRANCH.01, [C]; the gravity route is unconditional — red-team [`verification/redteam/rt_F_qft4d.py`]). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal  $c=8$  chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (SCALE.OVERDET.01).

## Summary

| Item            | Status  | Honest handle / what is missing                                                                                                                                                                                                                                                                                                                                          |
|-----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\eta_B$        | [C]     | downstream readout = $6.1 \times 10^{-10}$ from closed $\Omega_b h^2$ ; leptogenesis interface, cleanest scenario $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_{\Lambda}$ , $\tilde{m}_1 = m_3/A_{\Lambda}$ (shared decouple, no seesaw scale)                                                                                                                     |
| $m_p/m_e$       | [O]     | cross-sector (QCD scale / closed $m_e$ ); scheme-level, not a compiler power                                                                                                                                                                                                                                                                                             |
| Koide $Q$       | [C]     | source ratio $Q = 0.664$ near $2/3$ ; the source $\rightarrow$ pole factor $53/54$ is an <i>exact operator readout</i> of the missing $F=R+Q$ Sheet-Diamond corner [E]; the physical pole transfer ( $F_{\text{pole}}$ ) stays [C]                                                                                                                                       |
| dark matter     | [C]/[O] | determinant-line axion fixed: $f_a = M_{\text{scal}}/128$ , $m_a \approx 23.8 \mu\text{eV}$ ; the robust angle is the spine branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$ (mild-anharmonic, lands on $\Omega_{\text{DM}}$ untuned), an alternative to the over-producing $170^\circ$ hilltop — branch-level kill test, not a theory closure |
| muon $a_{\mu}$  | [C]     | seam vertex $a_{\mu}^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) = 2.879 \times 10^{-9}$ ; exact <i>value</i> , but the vertex projection is the [C] bridge ( $0.81\sigma$ dispersive, $\sim 1.5\sigma$ lattice)                                                                                                                                                  |
| quantum gravity | [E]/[O] | $R + R^2$ heat-kernel grounded (G2); physical sector gap-decoupled; only ambient limit (G6) open                                                                                                                                                                                                                                                                         |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**Net (honest).** Two of the five have a genuine quantitative handle that already lands on the data ( $\eta_B = 6.1 \times 10^{-10}$  via the closed  $\Omega_b$ ; the axion DM candidate with a closed misalignment angle). One is a computed *near-miss* reported without tuning (Koide 0.664 vs  $2/3$ ). One is structural-only and left open ( $m_p/m_e$  as a cross-sector ratio); quantum gravity has been sharpened — its  $R + R^2$  form is now heat-kernel grounded (G2) and its physical sector is gap-decoupled (G5), leaving only the ambient projective limit (G6) as a mathematical-completeness item. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

## Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is  $\Omega_b = (1 - \frac{1}{4\pi})\varphi_0^{\text{ret}}$  (whence the  $\eta_B$  order) in `verification/v7_gravity_cosmo.py`. The  $R + R^2$  structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); only the proton/electron ratio and the ambient projective limit (G6) are left open by design —

no script “proves” those, consistent with their **[O]** marking. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).